TOPOLOGICAL PRODUCTS UNDER BASE CHANGE BY A PRO-AFFINE SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma limits-lemma-topology-limit (canonical tag 01YY). cited at proof lines 1281-1283 and 1297. Topology of the inverse limit; used twice.

Source reference. Original lemma 01YY: sources/stacks-chow/tags.txt, source lines 2573–2573. Cited at proof lines 1281-1283 (for |T| = lim T_i) and line 1297 (for the product chain); load-bearing.

Lemma 0.2. *Let k be a field. Let $T = \mathrm{Spec}(A)$ where A is a directed colimit of algebras which are finite products of copies of k. For any scheme X over k we have $|T \times_k X| = |T| \times |X|$ as topological spaces.*

Proof. By taking an affine open covering we reduce to the case of an affine X. Say $X = \mathrm{Spec}(B)$. Write $A = \mathrm{colim}\, A_i$ with $A_i = \prod_{t \in T_i} k$ and T_i finite. Then $T_i = |\mathrm{Spec}(A_i)|$ with the discrete topology and the transition morphisms $A_i \to A_{i'}$ are given by set maps $T_{i'} \to T_i$. Thus $|T| = \lim T_i$ as a topological space, see Limits, Lemma 01YY. Similarly we have

$$\begin{aligned} |T \times_k X| &= |\mathrm{Spec}(A \otimes_k B)| \\ &= |\mathrm{Spec}(\mathrm{colim}\, A_i \otimes_k B)| \\ &= \lim |\mathrm{Spec}(A_i \otimes_k B)| \\ &= \lim |\mathrm{Spec}(\prod\nolimits_{t \in T_i} B)| \\ &= \lim T_i \times |X| \\ &= (\lim T_i) \times |X| \\ &= |T| \times |X| \end{aligned}$$

by the lemma above and the fact that limits commute with limits. □

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